

Positive Definiteness of K_d

Setup. Let $\mathbf{d} \in \mathbb{R}^n$ be the Lagrangian finger-space coordinates, $\mathbf{d}_{\text{ref}}$ the virtual equilibrium, and $\delta\mathbf{d} = \mathbf{d}_{\text{ref}} - \mathbf{d}$ the current deflection. The predicted tip force along contact normal $\mathbf{n}$ is

$$\mathbf{f} = \mathbf{n}^\top \mathbf{A} K_d \delta\mathbf{d}, \quad \mathbf{A} = [(\mathbf{P}\mathbf{J}_x)^*]^\top \eta \mathbf{J}_d^\top.$$

Define $\mathbf{v} = \mathbf{A}^\top \mathbf{n}$ (force-sensitivity direction in finger space) and $\mathbf{u} = \delta\mathbf{d}$ (current deflection), so that $\mathbf{f} = \mathbf{v}^\top K_d \mathbf{u}$. Set $\gamma = \|\mathbf{v}\|^2 \|\mathbf{u}\|^2$.

Definition 1. K_d is *positive definite* if $\mathbf{x}^\top K_d \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$, equivalently if its symmetric part $K_s := \frac{1}{2}(K_d + K_d^\top)$ is positive definite in the usual sense.

1. Force-tracking update

The loss $\mathcal{L} = \frac{1}{2}(\mathbf{f} - \mathbf{f}_{\text{des}})^2$ has gradient $\partial\mathcal{L}/\partial K_d = e \mathbf{v} \mathbf{u}^\top$ where $e = \mathbf{f} - \mathbf{f}_{\text{des}}$ is the force error. The update rule is

$$K_d^{(t+1)} = K_d^{(t)} - \alpha e_t \mathbf{v} \mathbf{u}^\top.$$

We assume $\mathbf{v}$ and $\mathbf{u}$ are fixed (quasi-static assumption).

Convergence and the role of α . Substituting the update into $\mathbf{f}^{(t+1)} = \mathbf{v}^\top K_d^{(t+1)} \mathbf{u}$:

$$\mathbf{f}^{(t+1)} = \mathbf{v}^\top K_d^{(t)} \mathbf{u} - \alpha e_t \underbrace{(\mathbf{v}^\top \mathbf{v})(\mathbf{u}^\top \mathbf{u})}_{=\gamma} = \mathbf{f}^{(t)} - \alpha \gamma e_t.$$

Therefore $e_{t+1} = (1 - \alpha\gamma) e_t$, a geometric sequence. For $e_t \rightarrow 0$ we need $|1 - \alpha\gamma| < 1$, i.e.

$$0 < \alpha < \frac{2}{\gamma}. \tag{1}$$

Under (1), $e_t = e_0(1 - \alpha\gamma)^t \rightarrow 0$ and the total accumulated update is

$$\alpha \sum_{t=0}^{\infty} e_t = \frac{\alpha e_0}{1 - (1 - \alpha\gamma)} = \frac{e_0}{\gamma}.$$

Theorem 1. Let $K_d^{(0)} \succ 0$, $\mathbf{f}_{\text{des}} > 0$, and α satisfy (1). Then $K_d^{(t)} \succ 0$ for all $t \geq 0$, for every such α , if and only if the initial force overshoot satisfies

$$e_0 = \mathbf{f}^{(0)} - \mathbf{f}_{\text{des}} < \lambda_{\min}(K_s^{(0)}) \cdot \|\mathbf{v}\| \|\mathbf{u}\|. \tag{2}$$

Proof. Let $\mathbf{P} = \frac{1}{2}(\mathbf{v} \mathbf{u}^\top + \mathbf{u} \mathbf{v}^\top)$ be the symmetric part of the rank-1 update direction. Each step changes K_s by $-\alpha e_t \mathbf{P}$. The largest eigenvalue of $\mathbf{P}$ satisfies $\lambda_+(\mathbf{P}) = (\mathbf{v}^\top \mathbf{u} + \|\mathbf{v}\| \|\mathbf{u}\|)/2 \leq \|\mathbf{v}\| \|\mathbf{u}\|$ (Cauchy-Schwarz), so Weyl's inequality gives

$$\lambda_{\min}(K_s^{(t+1)}) \geq \lambda_{\min}(K_s^{(t)}) - \alpha |e_t| \|\mathbf{v}\| \|\mathbf{u}\|.$$

Telescoping and substituting $\alpha \sum_t |e_t| = |e_0|/\gamma$ (derived above, independent of α):

$$\lambda_{\min}(K_s^{(T)}) \geq \lambda_{\min}(K_s^{(0)}) - \frac{|e_0|}{\|\mathbf{v}\| \|\mathbf{u}\|}, \quad \forall T.$$

This is strictly positive if and only if (2) holds. When $e_0 \leq 0$, K_s only increases so the result is immediate. $\square$

Key observation. Reducing α does not help: smaller steps are offset by more of them, and the total eigenvalue shift $\alpha \sum_t |e_t| \cdot \|\mathbf{v}\| \|\mathbf{u}\| = |e_0| / \|\mathbf{v}\| \|\mathbf{u}\|$ is the same for every α satisfying (1). If (2) fails, K_d eventually becomes indefinite regardless of α .

Lemma 1 (Isotropic initialization). *If $K_d^{(0)} = k_0 \mathbf{I}$ with $k_0 > 0$, condition (2) holds automatically and $K_d^{(t)} \succ 0$ for all $t \geq 0$ and all α satisfying (1).*

Proof. By Cauchy–Schwarz, $f^{(0)} = k_0 \mathbf{v}^\top \mathbf{u} \leq k_0 \|\mathbf{v}\| \|\mathbf{u}\|$, so $e_0 / (\|\mathbf{v}\| \|\mathbf{u}\|) \leq k_0 - f_{\text{des}} / (\|\mathbf{v}\| \|\mathbf{u}\|) < k_0 = \lambda_{\min}(K_s^{(0)})$. $\square$

2. Feedforward stiffness inversion

The first-order tip stiffness is the congruence transformation $K_x = \mathbf{M}^\top K_d \mathbf{M}$, $\mathbf{M} = \mathbf{J}_d(\mathbf{P} \mathbf{J}_x)^*$. Given $K_{x,\text{des}} \succ 0$ and invertible $\mathbf{M}$, inverting yields $K_d = (\mathbf{M}^\top)^{-1} K_{x,\text{des}} \mathbf{M}^{-1}$.

Theorem 2. $K_d = (\mathbf{M}^\top)^{-1} K_{x,\text{des}} \mathbf{M}^{-1} \succ 0$ *always*.

Proof. For any $\mathbf{x} \neq \mathbf{0}$, set $\mathbf{y} = \mathbf{M}^{-1} \mathbf{x} \neq \mathbf{0}$. Then $\mathbf{x}^\top K_d \mathbf{x} = \mathbf{y}^\top K_{x,\text{des}} \mathbf{y} > 0$. $\square$